



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Midterm– Semester FALL 2025

Course Title:	Digital System Design				Course Code:	CPE344	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais				Program Name:	BS Computer Engineering		
Semester:	7 th	Batch:	FA22-BCE	Section:	A & B	Date:	5 th November 2025	
Time Allowed:	90 Minutes				Maximum Marks:		50	
Student's Name:					Reg. No.			

Important Instructions/Guidelines:

- Write your Student's Name and Reg. No. clearly on the first page.
- The exam is closed book, closed notes.
- Mobile phones are not allowed in the examination room.
- Sharing of Calculator is not allowed. (Use your own calculator).
- Attempt all questions.

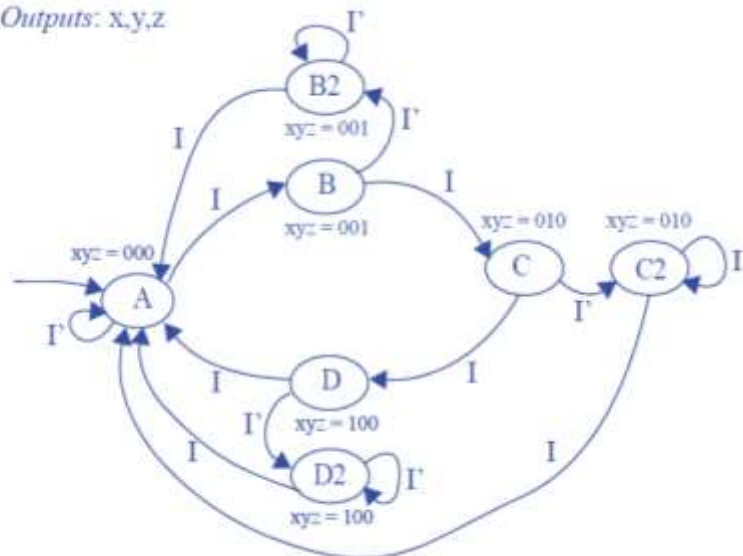
Question 1:

[CLO1-PLO3-C5]

(20 marks)

Draw a state diagram and *design* a controller for an FSM with an input I and three outputs x , y , and z . The output xyz should always exhibit the following sequence: 000, 001, 010, 100, repeat. The output should change on a rising clock edge when I is asserted. The input I can stop the sequence when set to 0. When I returns to 1, the sequence starts from 000 again. Make 000 the initial state.

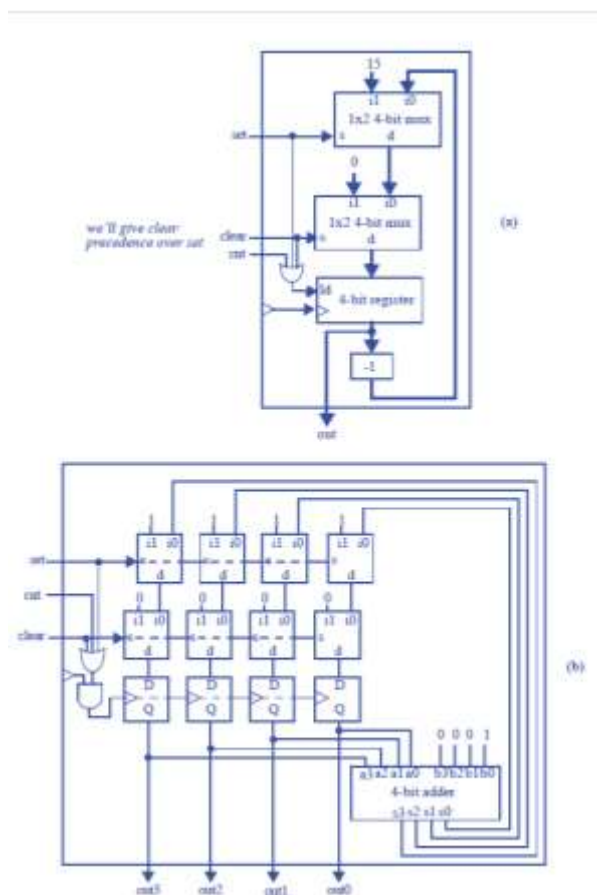
Inputs: I , Outputs: x, y, z



Step	Description
Step 1 <i>Capture the FSM</i>	Create an FSM that describes the desired behavior of the controller.
Step 2 <i>Create the architecture</i>	Create the standard architecture by using a state register of appropriate width, and combinational logic with inputs being the state register bits and the FSM inputs and outputs being the next state bits and the FSM outputs.
Step 3 <i>Encode the states</i>	Assign a unique binary number to each state. Each binary number representing a state is known as an <i>encoding</i> . Any encoding will do as long as each state has a unique encoding.
Step 4 <i>Create the state table</i>	Create a truth table for the combinational logic such that the logic will generate the correct FSM outputs and next state signals. Ordering the inputs with state bits first makes this truth table describe the state behavior, so the table is a state table.
Step 5 <i>Implement the combinational logic</i>	Implement the combinational logic using any method.

Question 2: [CLO1-PLO3-C5] (10 marks)

Design a 4-bit down-counter that has three control inputs: *cnt* enables counting up, *clear* synchronously resets the counter to all 0s, and *set* synchronously sets the counter to all 1s using flip-flops and muxes directly by following the register design process.



Question 3: [CLO2-PLO3-C5] (20 marks)

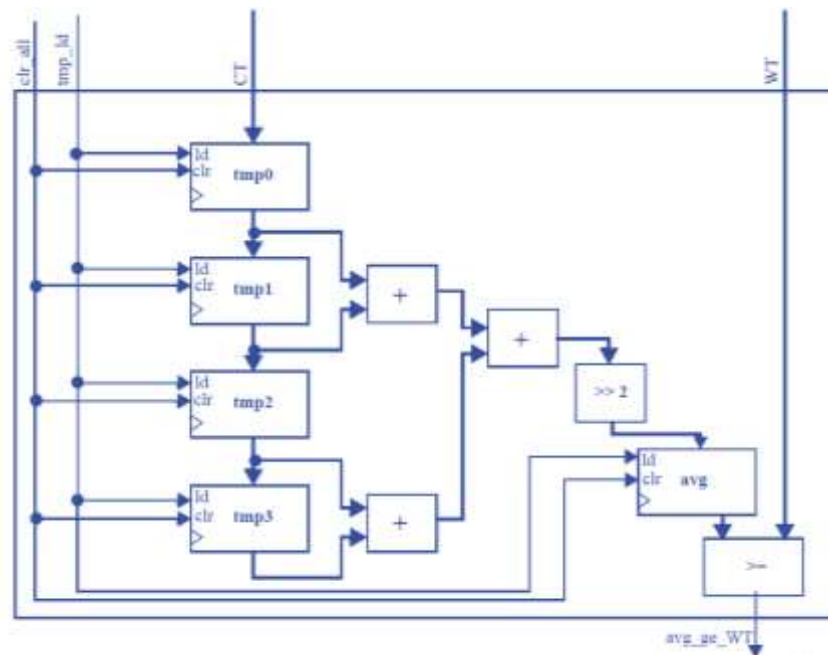
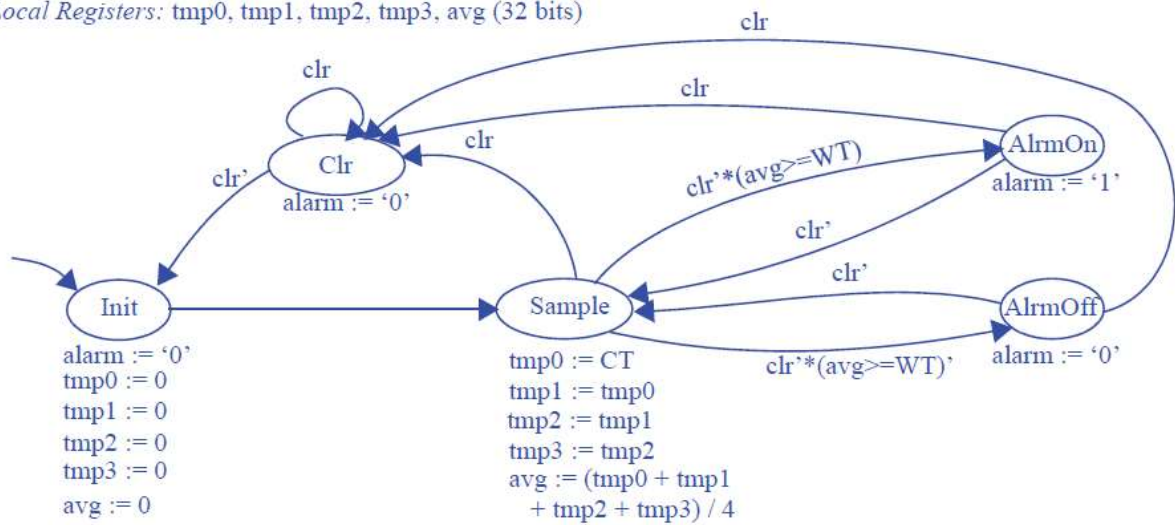
Design an alarm system that sets a single-bit output alarm to 1 when the average temperature of four consecutive samples meets or exceeds a user-defined threshold value. A 32-bit unsigned input *CT* indicates the current temperature, and a 32-bit unsigned input *WT* indicates the warning threshold.

Samples should be taken every few clock cycles. A single-bit input *clr* when 1 disables the alarm and the sampling process. Start by capturing the desired system behavior as an HLSM, and then convert to a controller/datapath by using the RTL design process.

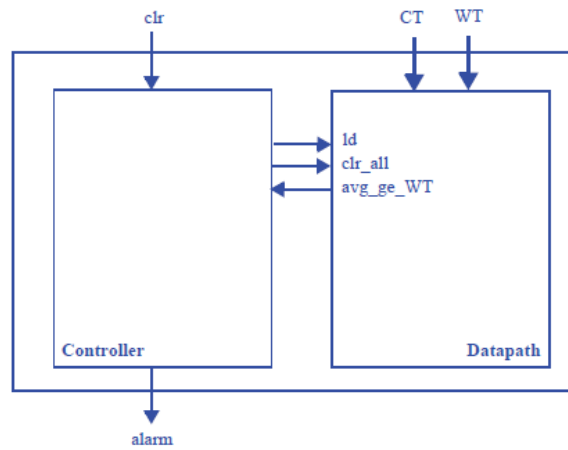
Inputs: CT, WT (32 bits); *clr* (bit)

Outputs: alarm (bit)

Local Registers: tmp0, tmp1, tmp2, tmp3, avg (32 bits)



Step 2B- Connect the datapath to a controller



Step 2C - Derive the controller's FSM

Inputs: `clr`, `avg_lt_WT`

Outputs: `alarm`, `clr_all`, `ld`

